

19 Linear Combinations of Random Variables

- In general, the distribution of a sum (or linear combination) of random variables can be difficult to obtain and depends on the joint distribution of the random variables.
- But there are some special cases where the distribution of a sum (or linear combination) works out nicely.
- And there are shortcut formulas that allow us to find the expected value and variance of a sum (or linear combination) without first finding its distribution.

Example 19.1. At a small hospital, let X be the number of babies born in a 5 day work week and let Y be the number of babies born on a weekend. Then $W = X + Y$ is the total number of babies born in a week. Assume that

- X and Y are independent
- X has a Poisson(2.1) distribution
- Y has a Poisson(0.4) distribution

Table 19.1 displays the marginal distributions of X and Y . (The probabilities have been rounded a little.)

Table 19.1: Distributions in Example 19.1

x, y	0	1	2	P($X = x$)
0				0.122
1				0.257
2				0.270
3				0.189
4				0.099
5				0.042
6				0.021
P($Y = y$)	0.670	0.268	0.062	

1. Create a table for the joint distribution of X and Y .

2. Compute and interpret $P(W = 0)$.
 3. Compute and interpret $P(W = 1)$.
 4. Compute and interpret $P(W = 2)$.
 5. Find the distribution of W .
 6. Compute and interpret $E(W)$. How does it relate to $E(X)$ and $E(Y)$?
 7. Compute $\text{Var}(W)$. How does it relate to $\text{Var}(X)$ and $\text{Var}(Y)$?
- **Poisson aggregation.** If X and Y are independent, X has a $\text{Poisson}(\mu_X)$ distribution, and Y has a $\text{Poisson}(\mu_Y)$ distribution, then $X + Y$ has a $\text{Poisson}(\mu_X + \mu_Y)$ distribution.
 - If component counts are independent and each has a Poisson distribution, then the total count also has a Poisson distribution.

Example 19.2. We'll illustrate some ideas with a small sample of simulated values. Say we have SAT Math scores (X) and Reading scores (Y) for 10 students. We're interested in the sum ($T = X + Y$) and difference (Math minus Reading, $D = X - Y$) of the scores.

The 10 X values are the same in each scenario, and the 10 Y values are the same in each scenario, but the (X, Y) values are paired in different ways:

- Scenario 1: correlation is 0.78
- Scenario 2: correlation is about 0 (actually -0.04)
- Scenario 3: correlation is -0.94

Table 19.2: Example 19.2 scenario 1, correlation is 0.78

Student	X	Y	T	D
1	520	530	1050	-10
2	540	470	1010	70
3	620	670	1290	-50
4	620	630	1250	-10
5	630	600	1230	30
6	670	610	1280	60
7	700	680	1380	20
8	700	580	1280	120
9	710	640	1350	70
10	760	720	1480	40
Mean	647	613	1260	34
SD	72	70	134	47
Corr(X, Y)	0.78			

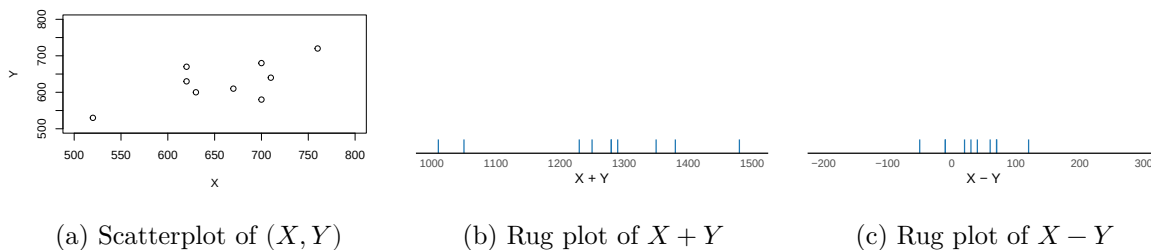


Figure 19.1: Example 19.2 scenario 1, correlation is 0.78

Table 19.3: Example 19.2 scenario 2, correlation is about 0

Student	X	Y	T	D
1	520	680	1200	-160
2	540	530	1070	10
3	620	630	1250	-10
4	620	600	1220	20
5	630	720	1350	-90
6	670	470	1140	200
7	700	670	1370	30
8	700	580	1280	120
9	710	640	1350	70
10	760	610	1370	150
Mean	647	613	1260	34
SD	72	70	98	103
Corr(X, Y)	-0.04			

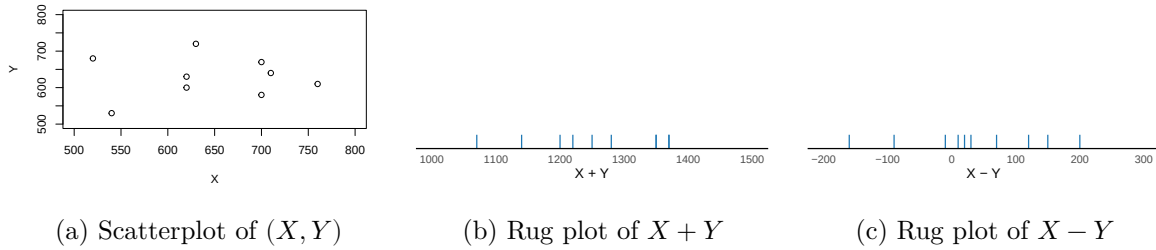
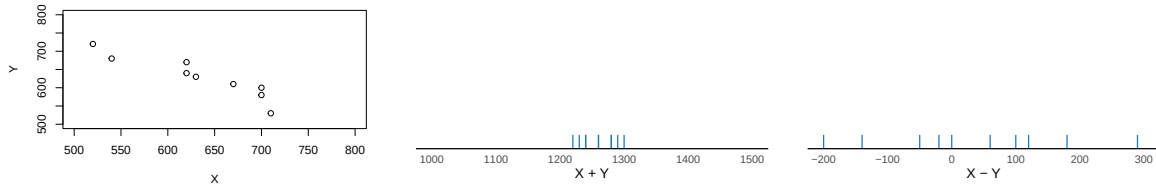


Figure 19.2: Example 19.2 scenario 2, correlation is about 0

Table 19.4: Example 19.2 scenario 3, correlation is -0.94

Student	X	Y	T	D
1	520	720	1240	-200
2	540	680	1220	-140
3	620	670	1290	-50
4	620	640	1260	-20
5	630	630	1260	0
6	670	610	1280	60
7	700	600	1300	100
8	700	580	1280	120
9	710	530	1240	180

Student	X	Y	T	D
10	760	470	1230	290
Mean	647	613	1260	34
SD	72	70	26	140
Corr(X, Y)	-0.94			



(a) Scatterplot of (X, Y)

(b) Rug plot of $X + Y$

(c) Rug plot of $X - Y$

Figure 19.3: Example 19.2 scenario 3, correlation is -0.94

In each scenario:

1. Compute the mean of total scores.
2. How does the mean of $X + Y$ relate to the means of X and Y ?
3. Compute the mean of the difference in scores (Math – Reading)
4. How does the mean of $X - Y$ relate to the means of X and Y ?

- **Linearity of expected value.** For *any* two random variables X and Y ,

$$E(X + Y) = E(X) + E(Y)$$

- That is, the expected value of the sum is the sum of expected values, regardless of how the random variables are related.
- Therefore, you only need to know the marginal distributions of X and Y to find the expected value of their sum. (But keep in mind that the *distribution* of $X + Y$ will depend on the joint distribution of X and Y .)
- Whether in the short run or the long run,

$$\text{Average of } X + Y = \text{Average of } X + \text{Average of } Y$$

regardless of the joint distribution of X and Y .

- A **linear combination** of two random variables X and Y is of the form $aX + bY$ where a and b are non-random constants. Combining properties of linear rescaling with linearity of expected value yields the expected value of a linear combination.

$$E(aX + bY) = aE(X) + bE(Y)$$

- For example ($a = 1, b = -1$): $E(X - Y) = E(X) - E(Y)$.
- Linearity of expected value extends naturally to more than two random variables.

$$E(a_1X_1 + \cdots + a_nX_n) = a_1E(X_1) + \cdots + a_nE(X_n)$$

Example 19.3. Continuing with the three scenarios in Example 19.2.

In each scenario:

1. Compute the variance of total scores.
2. How does correlation affect $\text{Var}(X + Y)$? In which of the three scenarios is $\text{Var}(X + Y)$ the largest? The smallest? Can you explain why? In which scenario is $\text{Var}(X + Y)$ roughly equal to the sum of $\text{Var}(X)$ and $\text{Var}(Y)$?

3. Compute the variance of the difference in scores (Math - Reading)

4. How does correlation affect $\text{Var}(X - Y)$? In which of the three scenarios is $\text{Var}(X - Y)$ the largest? The smallest? Can you explain why? In which scenario is $\text{Var}(X - Y)$ roughly equal to the *sum* of $\text{Var}(X)$ and $\text{Var}(Y)$?

- Variance of sums and differences of random variables

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X, Y)$$

- If X and Y are uncorrelated:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) \quad \text{if } X, Y \text{ are uncorrelated}$$

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) \quad \text{if } X, Y \text{ are uncorrelated}$$

- Variance of linear combinations: If a, b, c are non-random constants and X and Y are random variables then

$$\text{Var}(aX + bY + c) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X, Y)$$

Example 19.4. Let X be the sales of beer and Y the sales of salty snacks in a week at a certain supermarket, measured in thousands of dollars.

Suppose

- $E(X) = 22$, $\text{SD}(X) = 6$
- $E(Y) = 14$, $\text{SD}(Y) = 4$
- $\text{Corr}(X, Y) = 0.55$

1. Compute and interpret $E(X + Y)$.

2. Compute and interpret $SD(X + Y)$.

Example 19.5. Let X be the sales of beer and Y the sales of wine in a week at a certain supermarket, measured in thousands of dollars.

Suppose

- $E(X) = 22$, $SD(X) = 6$
- $E(Y) = 14$, $SD(Y) = 4$
- $\text{Corr}(X, Y) = -0.55$

1. Compute and interpret $E(X + Y)$.

2. Compute and interpret $SD(X + Y)$.

19.1 Exercises

Exercise 19.1. Consider a random variable X with $\text{Var}(X) = 1$. What is $\text{Var}(2X)$?

- Walt says: $SD(2X) = 2SD(X)$ so $\text{Var}(2X) = 2^2\text{Var}(X) = 4(1) = 4$.
- Jesse says: Variance of a sum is a sum of variances, so $\text{Var}(2X) = \text{Var}(X + X)$ which is equal to $\text{Var}(X) + \text{Var}(X) = 1 + 1 = 2$.

Who is correct? Why is the other wrong?

Exercise 19.2. Percent returns for assets X , Y , and Z follow a joint distribution with

- Mean 10 and standard deviation 15 for asset X ,
- Mean 5 and standard deviation 3 for asset Y ,
- Correlation of -0.6 between asset X and asset Y
- Asset Z yields a constant return of 1 percent.

An investment portfolio has 60% of its funds in asset X , 30% in asset Y , and 10% in asset Z .

1. Let R be the portfolio return. Express R in terms of X, Y, Z .
2. Compute $E(R)$.
3. Compute $\text{Cov}(X, Y)$.
4. Compute $\text{Cov}(X, Z)$.
5. Compute $\text{SD}(R)$.